Tahereh Fahi

AI PRODUCT ENGINEER

Toronto, Canada | Open to relocation to Tampa, FL

Email | LinkedIn | GitHub | Webpage

PROFILE

Co-Founder and AI Product Engineer delivering production GenAI, computer-vision, and statistical systems across fintech and neuroscience. Hands-on with Python and Go services, cloud deployment, data infrastructure, and model validation.

EDUCATION

MITx MicroMasters Program in Statistics and Data Science — Sep 2022 - Apr 2024
Massachusetts Institute of Technology (MIT) | **GPA: 4/4**

MSc - Information Technology Engineering (IT Systems) — Aug 2015 - May 2017
Tarbiat Modares University - Tehran, Iran | **Ranked 1st | GPA: 3.92/4**

BSc - Industrial Engineering (System Analysis) — Sep 2009 - Oct 2013
K.N. Toosi University of Technology - Tehran, Iran | **Ranked 1st | GPA: 3.35/4**

RESEARCH INTERESTS

- Biomedical Natural Language Processing
- Communication Dynamics in Multi-Agent LLM and Human-AI Systems
- Multi-Agent LLM Modeling
- AI Governance, Policy, and Community-Centered Design
- Information Diffusion, Misinformation Dynamics, and Network Effects
- Multi-Agent Modeling and Complexity in Sociotechnical Systems
- Health, Medical, and Bio-Informatics
- Computational Social Science and Information Dynamics
- Algorithmic Bias, Fairness, and Inequality in AI-Mediated Communication

PROFESSIONAL EXPERIENCE

Graduate Research Assistant — **Aug 2016 - Aug 2017**
Tarbiat Modares University - Tehran, Iran
Thesis: *"Integrating a Local Feature Selection and a Modified PLR Method for Stock Trading Points Prediction"* **GitHub**
- Tackled non-stationary, noisy financial time series by focusing on localized market behavior.
- Combined PLR-based time-series segmentation with Local Feature Selection to identify signals.

Co-Founder: AI Product Engineer — **Nov 2018 - Present**
RoboSelf - *Toronto, Canada*
- Designed and delivered production-ready, end-to-end applications using TypeScript, React/Next.js, Node.js/Python, PostgreSQL, REST APIs, and cloud infrastructure.
- Leveraged AI development tools such as ChatGPT, Codex, and Cursor to accelerate implementation, debugging, refactoring, testing, and technical documentation.
- Owned the complete software lifecycle, including system architecture, database design, API development, authentication, automated testing, CI/CD, Docker-based deployment, monitoring, and production support.
- Reviewed and validated all AI-assisted code to ensure security, scalability, performance, maintainability, and compliance with engineering standards.

TECHNICAL SKILLS

Languages: Python, R, MATLAB, MQL
Machine Learning Algorithms: Decision Tree, SVM, Linear Regression, Linear Programming, Clustering, Bayesian methods, Deep Learning, Gaussian Process, CNN, Reinforcement Learning (Q-Learning)
ML Technologies: TensorFlow, PyTorch, Keras, Pandas, scikit-learn, Matplotlib, Seaborn
Software Development Technologies: SQL, NoSQL
Big Data and Cloud Technologies: Microsoft Azure, Google Cloud
Computer Engineering: Data Structures, Algorithms, Object-Oriented Programming

PROJECTS

Deep Learning Specialization Projects

Generative Adversarial Networks for Realistic Image Synthesis
- Tackled the challenge of generating realistic synthetic imagery without labeled datasets.
- Built and trained PyTorch GAN models to improve adversarial stability.
- Achieved visually consistent outputs and enhanced feature diversity across samples.

Convolutional Neural Networks for Object Detection and Image Generation
- Addressed limitations of traditional classifiers in handling image spatial hierarchies.
- Designed CNN architectures with transfer learning and residual blocks.
- Improved accuracy and generalization through YOLO-based detection.

Deep Learning Specialization Projects (continued)

Sequence Modeling with RNNs and Attention Mechanisms for Natural Language Processing

- Confronted the difficulty of modeling sequential linguistic dependencies in text.
- Trained RNN, LSTM, and attention-based models for translation, sentiment, and trigger-word tasks.

Deep Learning Approaches for Emotion Recognition in Social Media Text

- Sought to detect emotional tone and polarity in informal, high-noise tweet data.
- Engineered tokenization and embedding layers with deep classifiers for multi-label emotion prediction.
- Produced robust classification results that outperformed classical sentiment baselines.

Optimization and Regularization Strategies for Neural Network Training

- Faced issues of slow convergence and overfitting in deep model training.
- Applied initialization schemes, dropout, L2 regularization, and batch normalization in TensorFlow.

Bias-Variance Trade-Off and Transfer Learning Strategies in Machine Learning

- Identified challenges of model underfitting, overfitting, and data mismatch in ML pipelines.
- Diagnosed errors and optimized architectures using transfer learning.

Implementation of Deep Neural Networks for Structured and Image Data

- Built neural networks from scratch to understand backpropagation mechanics.
- Verified correctness through gradient checks.

MIT MicroMasters Projects

Q-Learning with Function Approximation for Sparse-Reward Text-Based Games (GitHub)

- Aimed to teach an agent to learn optimal policies under sparse-reward conditions.
- Implemented Q-learning with linear approximation and adaptive exploration.
- Achieved stable convergence and measurable improvement in cumulative reward rates.

Progressive Digit Recognition on MNIST: From Linear Models to CNN

- Investigated performance limits of linear classifiers on handwritten digits.
- Constructed a pipeline from logistic regression to CNN architectures with dropout.
- Attained more than 98% accuracy while improving model robustness and computational efficiency.

Dimensionality Reduction and Classification of Single-Cell RNA-Seq Data (GitHub)

- Addressed the problem of analyzing noisy, high-dimensional genomic datasets.
- Applied PCA and t-SNE for dimensionality reduction and clustered cells.
- Discovered distinct cell types with cross-validation metrics.

Comparative Graph Analysis of Facebook and Twitter Networks Using Structural and Statistical Models (GitHub)

- Explored assortativity and topological differences between social networks.
- Computed degree distributions, centralities, and power-law fits to evaluate network models.

Graph-Based Centrality and Community Detection in Criminal and Citation Networks

- Tackled identification of key actors and clusters within complex networks.
- Applied centrality metrics and community-detection algorithms on graph datasets.
- Revealed critical nodes and relationships informing intervention.

Collaborative Filtering with Gaussian Mixture Models for Sparse Movie Rating Prediction

- Confronted the problem of missing user ratings in recommendation systems.
- Modeled latent preferences using Gaussian mixture models with expectation-maximization.
- Improved prediction consistency and reduced BIC-based model penalty under sparse data.

MIT MicroMasters Projects (continued)

Forecasting CO2 Concentrations and Economic Indicators Using ARIMA Models *(**GitHub**)*

- Faced non-stationary patterns in environmental and economic time series.
- Built ARIMA models with residual diagnostics and forecast validation.

CPI and BER Inflation Data Analysis with External Regressors and Model Improvements *(**GitHub**)*

- Investigated the relationship between CPI inflation and BER expectations for forecasting.
- Integrated external regressors into ARIMAX models and tested cross-correlation effects.
- Enhanced predictive accuracy and model stability across validation horizons.

Gaussian Process Models for Spatial Prediction of Environmental Data *(**GitHub**)*

- Aimed to predict spatial variables such as ocean currents and temperature fields.
- Built Gaussian Process models with custom covariance kernels and interpolated flow patterns from Philippine Archipelago data.

TensorFlow Course Projects

- Design and Optimization of Multilayer Feedforward Neural Architectures Using TensorFlow
- Advanced Convolutional Frameworks for Image Recognition in TensorFlow
- Leveraging Pretrained Deep Models for Domain-Specific Applications via Transfer Learning

CERTIFICATIONS AND COURSEWORK

Related Extra Courses

Machine Learning on Google Cloud - Coursera 2020
Deep Learning Specialization - Coursera - DeepLearning.AI 2020
Machine Learning - Coursera - Stanford University 2024

Related Attended Courses

Graduate Courses

- Machine Learning
- Data Mining and Knowledge Discovery
- Information and IT
- Web Architecture and Programming

Undergraduate Courses

- Systems Analysis
- Computer Programming
- Engineering Statistics
- Computer Applications in Industrial Engineering

AWARDS

- Ranked 99th among approximately 32,000 participants in the Information Technology Engineering National M.Sc. Entrance Examination for universities of Iran 2015
- Ranked among the top 7% of more than 260,000 participants in the Iranian National Universities Entrance Examination for the B.Sc. degree 2009
- Selected as a qualified participant in the first stage of the Iranian National Computer Olympiad 2006

LANGUAGE PROFICIENCY

English: IELTS 7.5 (Listening 8.0, Speaking 7.5, Reading 7.0, Writing 6.5)

REFERENCES

Dr. Seyed Kamal Chaharsooghi, Professor at Tarbiat Modares University, Department of Industrial and Systems Engineering, Tehran, Iran
Email: skch@modares.ac.ir

Dr. Milad Jasemi, Professor at the University of Montevallo and Assistant Professor of Data Analytics/Department Chair
Email: emjasemiz@montevallo.edu